

SPL

c) The concept of an independent judiciary was taken from America.

12/11/25

Q) Briefly discuss the features of Indian Constitution.

A- * Three organs

- There are three organs of the govt.

a) Legislature - It makes the laws

b) Executive - It implements the laws.

c) Judiciary - It interprets the laws.

It is based on the principle of 'Separation of Powers' which is essential to their effective working.

* Three levels of the govt.

In India, there is 3 tier system of governments.

Central Govt.

State Govt.

Local Self Govt.

- Some functions and responsibilities are reserved for Central Govt. and others are assigned to the state govt.

* Fundamental Rights

- Some fundamental rights are provided to the Indian citizens by the Constitution. They are as follows -

a) Right to Equality (Article 14)

b) Right to freedom (Article 21)

c) Right against exploitation
(Article 23)

etc.

* Directive principles of State Policy

- Directive Principles are goals that the Govt. should try to achieve. They reflect the direction to make India better. For eg.

a) Article 41 - Welfare govt.

b) Article 44 - Uniform civil
code

c) Article 38 - Social, Political
& Economical justice

d) Article 48 A - Environment
& Wild life protection

e) Article 49 - Protection of monuments, palaces, etc.

f) Article 47 - Nutrition & Public Health

* Fundamental Duties

MATH

men req. to reap the field in
 30 days = 16 men

men req. to reap the field in
 1 day = (16×30) days.

$\therefore$ men req. to reap the field in
 24 days = $(\cancel{16} \times \cancel{30}) \times \frac{24}{30} = 20 \text{ men}$

$$\begin{array}{r} 24 \\ -12 \\ \hline 631 \end{array}$$

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Q) In which of the following table
 a & b vary inversely.

(i)

a	8	16	32	256
b	32	16	8	1

(ii)

a	9	18	2	12
b	8	4	30	6

$\therefore$ We know that a & b vary
 inversely if the product of
 a & b remains same for
 all values of a & b.

Now,

$$(i) \quad 8 \times 32 = 256;$$

$$16 \times 16 = 256;$$

$$32 \times 8 = 256;$$

$$256 \times 1 = 256$$

We see that the product of ab in all the cases is constant so a & b (i) vary inversely.

Now,

$$(ii) \quad 9 \times 8 = 72;$$

$$18 \times 4 = 72;$$

$$30 \times 2 = 60;$$

$$12 \times 6 = 72$$

We see that product of ab is not constant so a & b (ii) do not vary ~~in~~versely.

Q) If 52 men can do a piece of work in 35 days in how many days 28 men will do it?

Let 28 men do the work in x days.

No. of men	52	28
No. of days	35	x

Clearly less is the no. of men more will be the no. of days, so it is a case of inverse variation.

ATQ,

$$52 : 28 = x : 35$$

$$\Rightarrow \frac{52}{28} = \frac{x}{35}$$

$$\Rightarrow 28x = 52 \times 35$$

$$\Rightarrow x = \frac{52 \times 35}{28} = 65$$

Hence 28 men will take 65 days.

Q) If 56 men can do a piece of work in 42 days. How many men will do it in 14 days.

Let the no. of men be u .

No. of men	56	u
No. of days	42	14

[less days, more men $\therefore$ inverse variation]

ATQ,

$$56 : u = 42 : 14$$

$$\Rightarrow \frac{56}{u} = \frac{42}{14}$$

$$\Rightarrow 42u = 56 \times 14$$

$$\Rightarrow u = \frac{56 \times 14}{42} = \frac{56}{3}$$

- Q) 1000 soldiers in a fort had enough food for 20 days but some soldiers were transferred to another fort and the food lasted for 25 days. How many soldiers were transferred?

Let the No. of soldiers ~~transferred~~ ~~be~~ be in the fort after some soldiers transferred.

No. of soldiers	1000	x
food lasted for	20	25

ATQ,

$$1000 : x = 25 : 20$$

$$\Rightarrow \frac{1000}{x} = \frac{25}{20}$$

$$\Rightarrow 25x = 1000 \times 20$$

$$\Rightarrow x = \frac{1000 \times 20}{25} = 800$$

Q) If u & y vary inversely as each other & $u = 10$, $y = 6$ find y when $u = 15$.

$\therefore u$ & y vary inversely
 $\therefore u y$ remains constant

u	10	15
y	6	y

Now,

$$10 \times 6 = 60$$

$$\therefore 15 \times y = 60$$

$$\Rightarrow y = \frac{60}{15} = 4$$

$$\therefore y = 4$$